

LEONID ELKIN

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Education

University of California San Diego

Bachelor of Science in Artificial Intelligence

Expected June 2030

San Diego, CA

Tonbridge School

*A Levels: Mathematics, Further Mathematics, Computer Science, Physics (A*A*A*A); EPQ (A*)*

June 2026

Kent, UK

- Honorary Scholar; 11 Grade 9s at GCSE, a top 150 result nationally

Projects

Drone Strike Map | *Python, Flask, SQLite, Leaflet, nginx, systemd, Claude API*

June 2026 – Present

- Launched a war-tracking map with **2,000+ monthly users** charting every reported drone and missile strike
- Built scrapers for **51 sources** in 3 languages, run every 30 minutes with rule-based and LLM extraction
- Published a public REST API (13 endpoints), served from a Linux server behind nginx with TLS and rate limiting

Yagi-Uda Radar & CanSat | *Raspberry Pi, RF transceivers, antenna design*

Sept 2024 – Present

- Led a 7-person UK Space Agency CanSat team, building the payload and radio link for a satellite that flew
- Built a 20 km-range handheld SSR radar from scratch around a 14.5 dBi Yagi-Uda antenna for tracking aerial targets

Penumbra | *Python, PyQt6, OpenGL, TCP sockets, PyInstaller*

July 2026 – Present

- Developed a real-time naval strategy game (**16,000+ lines**) with 12 boss levels, 20+ unit types and AI opponents
- Wrote a relay server for online multiplayer that pairs players by room code, with no port forwarding

MoveGrade | *JavaScript, Chrome Extensions (MV3), WebAssembly, Stockfish*

June 2026 – Aug 2026

- Built a Chrome extension that grades every chess move live on lichess and chess.com, running Stockfish in WebAssembly and judging openings against a book of 3,328 named lines

Yavalath & Pentalath | *Python, A Level coursework*

Sept 2024 – June 2026

- Developed two hex strategy board games solo (**7,000+ lines**) with sound and a computer opponent

Research

Globular Cluster Lifespan | *Python, NumPy; paper and poster*

Feb 2025 – June 2025

- Built an N-body simulation measuring how primordial binary stars affect the lifespan of globular clusters

Neural Scaling Laws | *Python, NumPy, LaTeX; Extended Project Qualification, A**

Jan 2025 – May 2025

- Implemented neural networks from scratch in NumPy and trained them to see what effect hyperparameter tweaking has on the accuracy and compute usage (56-page paper)
- Published the library on PyPI as **elkwork**, reaching 98.5% on MNIST and 93.4% on Fashion-MNIST

Technical Skills

Languages: Python, JavaScript, Java, HTML/CSS, SQL, PowerShell

Machine Learning: NumPy, neural networks from scratch, training and evaluation, computer vision, LLM APIs

Web & Systems: Flask, REST APIs, SQLite, Linux, nginx, systemd, Chrome Extensions, WebAssembly, Git

Other: PyQt, pygame, LaTeX, Raspberry Pi, RF and antenna design

Leadership / Extracurricular

3D Printing Society

Sept 2025 – July 2026

Founder

Tonbridge School

.22 Rifle Shooting

May 2024 – July 2026

UK National U21 Team; Division One Junior Team

Programming Society

Sept 2025 – June 2026

Head

Tonbridge School

- Lectured audiences of 40+ students on fundamental machine learning architectures